Breast Cancer Detection Using Ultrasound Image Classification

Andronescu
Biomedical Engineering
Stevens Institute of Technology
Hoboken, USA
mandrone@stevens.edu

Gauthier
Computer Science
Stevens Institute of Technology
Hoboken, USA
sgauthie1@stevens.edu

Bermeo
Computer Science
Stevens Institute of Technology
Hoboken, USA
nbermeo@stevens.edu

***Abstract*—Breast cancer is one of the most common cancers affecting women and early detection is important for improving patient outcomes. In this project, we developed a convolutional neural network from scratch and compared it to ResNet18 to classify ultrasound images as malignant or non-malignant. The dataset contained ultrasound images labeled as malignant, benign, and normal, and we combined benign and normal into non-malignant for binary classification. The images were split into training, validation, and test sets to then be evaluated with the metrics: accuracy, precision, recall, F1-score, and a confusion matrix. Our results show that the models were able to learn meaningful information, but RestNet18 was more accurate. These findings suggest that deep learning can support breast cancer image classification in an actual clinical environment to work side by side with doctors one day, although additional data, tuning, and clinical validation would be needed.**

***Keywords*— Breast cancer detection, ultrasound imaging, convolutional neural network, RestNet18, medical image classification, deep learning, PyTorch**

I. Background

As of April 2026, breast cancer is the most common form of cancer in U.S. women, accounting for about 32% of newly diagnosed cancers in women [1]. With these rates, it is predicted that 42,140 U.S. women will die due to it in 2026. Depending on the race and ethnicity of the individual, specific patients have higher risks of death – specifically those who are black and male. The mortality rate is strongly correlated with the time of diagnosis, as successful treatment for breast cancer occurs at earlier stages of cancer progression [2]. Therefore, it is vital for patients, especially those in marginalized communities, to be able to accurately identify and predict the development of breast cancer in a timely manner.

Breast cancer is diagnosed by performing various tests including mammograms, breast MRIs, core needle biopsies, breast exams, and breast ultrasounds [3]. Of these, breast ultrasounds have been shown to be the most effective at diagnosing breast cancer while being one of the least invasive and low-cost diagnostic methods.

In an era of AI applications in medicine, deep learning has been a useful tool to better identify and predict various medical conditions such as breast cancer. Modern breast cancer machine learning models have often utilized breast ultrasound datasets, as breast ultrasounds have shown to have the highest accuracy in breast cancer classification, likely due to the greater size of ultrasound datasets from a wider range of patients despite economic statuses [2,4].

The most common and effective machine learning models for breast cancer prediction are convolutional neural networks (CNNs). CNNs are highly applicable towards breast cancer diagnosis because they can identify subtle patterns in datasets, having high sensitivity and specificity for classifications in medical imaging [5]. CNNs are also capable of extracting features from noisy and low-contrast images in ultrasound images [6]. However, CNNs are known to have a tendency to underfit complex patterns.

CNNs designed for breast cancer diagnosis usually have a linear design and a 14-layer network [7]. Since datasets tend to have a small size, model performance is often limited by the number of examples within each category [8].

As CNNs are designed to be shallow to mitigate overfitting, they are often not capable of being accurate for a more diverse and complex dataset. ResNet 18 is a solution to this issue. It is an 18-layer residual network – still a type of CNN – which uses residual (skip) connections to avoid underfitting and vanishing gradients, where gradients become too small to allow for effecting learning at deeper levels [7,9]. To achieve this, residual blocks (basic blocks) are used to learn residual mapping rather than the original mapping, which simplifies training parameters: $F(x) = H(x) - x$ instead of $H(x)$ [9]. Compared with standard CNNs which prioritize specific features of images, when ResNet 18 observes an image, it has a more distributed heat map allowing it to identify tumors more easily [7].

It is vital that machine learning models have high diagnostic accuracy and precision when diagnosing medical issues, such as breast cancer. If there is a high rate of false positives, patients could have a greater risk of anxiety, unnecessary follow-ups, and invasive diagnostic procedures, which can be costly, especially for low-income patients. However, if the model has a high rate of false negatives, the patient's breast cancer could worsen, which would make treatment in the future less effective and more physically and emotionally harmful.

The aim of this project is to identify the differences and potential benefits of using a ResNet model as opposed to a standard CNN for classifying breast cancer ultrasound images. To facilitate this, CNN was prepared from scratch, and its diagnostic ability was compared with that of a pretrained ResNet 18 model. Due to the higher risk for the patient for false negatives, this project focuses on the recall or sensitivity, to prioritize true positive diagnosis.

II. METHOD

A. Dataset

The dataset used for this study is called the "Dataset of breast ultrasound images" from Al-Dhabyani, Geomaa, Khaled and Fahmy from Cario University and the National Cancer Institute [10]. Included in the dataset were 780 ultrasound images among women in ages between 25-75 years old collected in 2018. The images were classified into 3 classes: normal, benign (non-cancerous, but abnormal) and malignant with the respective numbers being 133, 487, and 210. The dataset included masks for every image, but we decided to train our models on the raw images since masks are created by experts and used for research/annotation. In a practical application, a model would help doctors predict malignancy using raw images. In addition, the masks themselves are made through expert validation and deep segmentation analysis. Having the model train on them wouldn't make sense as it'd have obvious signs for malignancy.

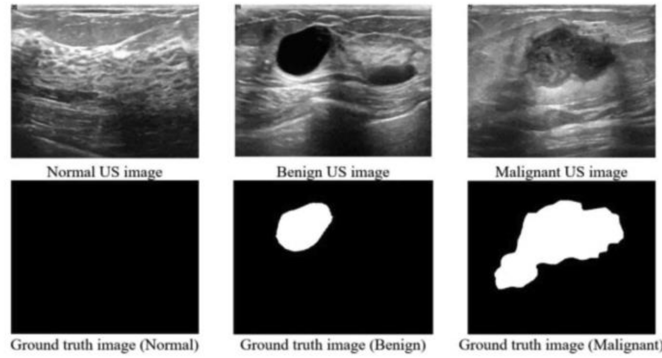

Fig. 1. Dataset Images

B. Program Structure

The structure of our program from our GitHub is process the data, create models, train models, evaluate 1st model, and evaluate 2nd model [11]. The preprocessing organized the data into separate folders for the raw image classes by collecting images from the malignant, benign, and normal folders, verified that each file was valid, and copied the files into the processed dataset directory. The processed dataset was divided into training, validation, and testing folders each containing malignant and non-malignant subfolders. In addition, the preprocessing pipeline converted the original three class dataset into a two-class dataset where benign and normal were placed into the non-malignant category. The dataset was then randomly split into training, validation and tests sets at a ratio of 70, 15, 15. In addition to these images, because the dataset was small and imbalanced, data augmentation was used during training with random flips and rotations to help increase variation and reduce overfitting.

The model architecture was the basic CNN and RestNet18. The basic convolutional neural network was implemented using PyTorch. The model consisted of convolutional layers, ReLU activation functions, pooling layers and fully connected layers for classification. The convolutional layers were used to extract the image features like edges, textures, and eventually the whole image, while the pooling layers reduced spatial dimensions and helped the model focus on important patterns. The final classifier produced, of course, the binary prediction. The ResNet18 on the other hand was pre-trained on millions of images from ImageNet and then fine tuned with our images in its last layer.

```python
class BasicCNN(nn.Module):
    def __init__(self, num_classes=2):
        super().__init__()

        self.features = nn.Sequential(
            # Layer 1 -
            nn.Conv2d(3, 32, kernel_size=3, padding=1),
            nn.ReLU(),
            nn.MaxPool2d(2),

            # Layer 2 -
            nn.Conv2d(32, 64, kernel_size=3, padding=1),
            nn.ReLU(),
            nn.MaxPool2d(2),

            # Layer 3 -
            nn.Conv2d(64, 128, kernel_size=3, padding=1),
            nn.ReLU(),
            nn.MaxPool2d(2),
        )

        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(128 * 28 * 28, 256),
            nn.ReLU(),
            nn.Dropout(0.5),
            nn.Linear(256, num_classes),
        )

    def forward(self, x):
        x = self.features(x)
        x = self.classifier(x)
        return x
```

Fig. 2. Basic CNN Architecture

Finally, there was a final file for training. The model was trained using the processed training set directories and then evaluated on the validation set after each epoch to ensure the model was actually learning instead of memorizing. The training file ran 10 epochs at a learning rate of 0.001 with Adam to compute the weights and the best model was saved based on validation performance.

Results

We used precision, recall, F1-score and a confusion matrix for both non-malignant and malignant for each model to evaluate which model performed better.

```
Classification Report:
               precision    recall  f1-score   support

    malignant       0.70      0.48      0.57        33
non_malignant       0.83      0.92      0.87        89

     accuracy                           0.80       122
    macro avg       0.76      0.70      0.72       122
 weighted avg       0.79      0.80      0.79       122

Confusion Matrix:
[[16 17]
 [ 7 82]]

Accuracy: 0.8032786885245902
```

Fig. 3. Basic CNN Metrics

Upon inspection of an accuracy of 80%, you may assume that the model is pretty good for our basic CNN. However, accuracy cannot tell the whole story in an imbalanced dataset. Our data is imbalanced since we have 211 malignant images and 587 non-malignant images which gives us a ratio of 1:2.8. Accuracy is not the full picture here because if the model just predicts non-malignant every time it would have an accuracy of around 75%. The most important metric to look at is recall which tells us more about the false negatives. A false negative is the most important metric to keep low in medical diagnosis because we do not want to tell a patient they do not have cancer when, in reality, they do which will lead to worsening health. The recall in our basic CNN for malignancy was not good at 48% meaning in all the actual positive cases we missed about half of them. Our precision for malignancy was 70% and the harmonic mean of both of them, the F1-score, was 57% giving our model a lackluster score. As for non-malignancy, the precision, recall and F1-scores were all above 80 which is good, but the model could have been biased toward non-malignant. In our confusion matrix for the testing data, we can further highlight false negatives with type 2 errors as 17 were found.

```
Classification Report:
               precision    recall  f1-score   support

    malignant       0.82      0.70      0.75        33
non_malignant       0.89      0.94      0.92        89

     accuracy                           0.88       122
    macro avg       0.86      0.82      0.84       122
 weighted avg       0.87      0.88      0.87       122

Confusion Matrix:
[[23 10]
 [ 5 84]]

Accuracy: 0.8770491803278688
```

Fig. 4. ResNet18 Metrics

As for our ResNet18 model, we found significantly higher metrics. For malignancy, the recall was 70% with a much higher score than the previous model and a F1 score of 75%. The non-malignancy performed better as well with metrics exceeding a score of around 90%. And the confusion matrix shows 7 less type 2 errors than before.

Comparing the two we can see that ResNet outperformed the basic model very well as the basic CNN missed many cancer cases and the ResNet18 missed less with more accuracy. The basic CNN provides a good baseline, but ResNet18 is stronger for this task in every metric.

Conclusion

In our study, we evaluated the efficiency of deep learning architectures for the binary classification of breast ultrasound images into malignant and non-malignant categories. Our results demonstrated a clear performance gap between the custom basic CNN versus the pre-trained ResNet18 model. While the scratch CNN managed to achieve an 80% accuracy, its low recall of 48% for the malignant cases presents a significant clinical risk in terms of false negatives (where patients who have malignant tumors are diagnosed as non-malignant). In comparison, by leveraging transfer learning and the architectural advantages of skip connections, the ResNet18 model achieved a higher accuracy of 88% and a significantly improved recall of 70%. These findings emphasize that, in a medical diagnostic context, recall is the most critical metric as opposed to raw accuracy. The goal is to minimize missed diagnoses, which could potentially be lifesaving. Overall, our project serves as a proof of concept that, while deep learning can capture biological patterns in noisy data, pre-trained architectures like the ResNet18 are significantly more robust than simpler linear models for complex medical imaging tasks.

Further Work

To bridge the gap between our experiment and clinical reliability, several areas for future research should be explored. In the case of our project, implementing extensive data augmentation (such as rotations, flipping, and scaling) could be effective for addressing the small sample size and reducing the risk of overfitting. Future versions should also focus on rebalancing the dataset to avoid the current majority-class bias risk. This could be done through oversampling of the malignant cases, for instance. Additionally, incorporating more advanced architectures (beyond the ResNet18) would allow for better interpretability and ensure that the model's predictions are based on relevant image features rather than imaging noise. For any potential clinical applications, future research should focus on developing more complex architectures capable of providing higher recall rates within medical environments. Ultimately, this project serves as a foundation for a future where AI works together with doctors to improve patient outcomes in breast cancer screening.

References

[1] Mayo Clinic, "Breast cancer," *Mayo Clinic*, Jan. 10, 2025. https://www.mayoclinic.org/diseases-conditions/breast-cancer/symptoms-causes/syc-20352470

[2] T. T. Yıldırım, O. Yaman, İ. Kılıç, B. Taşar, E. S. Timurkaan, and N. Aydoğdu, "Multi-Class Classification of Breast Ultrasound Images Using Vision Transformer-Based Ensemble Learning," *Diagnostics*, vol. 15, no. 17, p. 2235, Sep. 2025, doi: https://doi.org/10.3390/diagnostics15172235.

[3] Mayo Clinic, "Breast Cancer - Diagnosis and Treatment - Mayo Clinic," *Mayoclinic.org*, Jan. 10, 2025. https://www.mayoclinic.org/diseases-conditions/breast-cancer/diagnosis-treatment/drc-20352475

[4] K. Jabeen *et al.*, "Breast Cancer Classification from Ultrasound Images Using Probability-Based Optimal Deep Learning Feature Fusion," *National Library of Medicine*, Jan. 21, 2022. https://pmc.ncbi.nlm.nih.gov/articles/PMC8840464/ (accessed May 11, 2026).

[5] Y.-M. Wang *et al.*, "CNN-Based Cross-Modality Fusion for Enhanced Breast Cancer Detection Using Mammography and Ultrasound," *National Library of Medicine*, Dec. 12, 2024. https://pmc.ncbi.nlm.nih.gov/articles/PMC11679931/ (accessed May 11, 2026).

[6] S. Tiwari, G. Jain, D. K. Shetty, M. Sudhi, J. M. Balakrishnan, and S. R. Bhatta, "A Comprehensive Review on the Application of 3D Convolutional Neural Networks in Medical Imaging," *Engineering Proceedings*, vol. 59, no. 1, p. 3, 2023, doi: https://doi.org/10.3390/engproc2023059003.

[7] L. Reginato, "A Practical Comparison Between CNN and ResNet Architectures: A Focus on Attention Mechanisms," *Medium*, Oct. 14, 2024. https://medium.com/@leonardofonseca.r/a-practical-comparison-between-cnn-and-resnet-architectures-a-focus-on-attention-mechanisms-cee7ec8eca55

[8] Z. Zhu, S. Wang, and Y. Zhang, "A Survey of Convolutional Neural Network in Breast Cancer," *Cmes-computer Modeling in Engineering & Sciences*, vol. 136, no. 3, pp. 2127–2172, Jan. 2023, doi: https://doi.org/10.32604/cmes.2023.025484.

[9] "7.6. Residual Networks (ResNet) — Dive into Deep Learning 0.14.3 documentation," *d2l.ai*. https://d2l.ai/chapter_convolutional-modern/resnet.html

[10] Al-Dhabyani, W., Gomaa, M., Khaled, H., Fahmy, A., 2020. Dataset of breast ultrasound images. Data Brief 28, 104863.

[11] N. Bermeo, "BME571_BreastCancerCNN," GitHub repository, 2026. [Online]. Available: https://github.com/NelsonBermeo/BME571_BreastCancerCNN